

Solubility measurement and modeling for the system propane–water from 277.62 to 368.16 K

Antonin Chapoy^a, Salim Mokraoui^a, Alain Valtz^a, Dominique Richon^{a,*},
Amir H. Mohammadi^b, Bahman Tohidi^b

^a Ecole Nationale Supérieure des Mines de Paris, CENERG/TEP 35, Rue Saint Honoré, 77305 Fontainebleau, France

^b Centre for Gas Hydrate Research, Institute of Petroleum Engineering, Heriot-Watt University, Edinburgh EH14 4AS, Scotland, UK

Received 31 May 2004; accepted 27 August 2004

Available online 11 November 2004

Abstract

In this communication, a total of 61 new experimental data on the solubility of propane in water are reported over a wide temperature range (i.e., 277.62–368.16 K) up to 3.915 MPa (i.e., liquid water–vapor region). A static-analytic apparatus taking advantage of a pneumatic capillary sampler is used for fluid sampling. The Valderrama modification of the Patel–Teja equation of state combining with non-density dependent mixing rules is used for modeling vapor–liquid equilibrium. The new solubility data generated in this work are used for adjusting the binary interaction parameters between propane and water. The experimental data generated in this work are compared to literature data and to the calculations through the thermodynamic model tuned on our experimental data. In order to further evaluate the performance of the thermodynamic model, a comparison is made between the predicted water content data and the experimental data reported in the literature. The predictions are in good agreement with the independent experimental data, demonstrating the reliability of the thermodynamic approach used in this work.

© 2004 Elsevier B.V. All rights reserved.

Keywords: Vapor–liquid equilibria; Propane; Water; Experimental measurements; Thermodynamic model

1. Introduction

Accurate knowledge on the phase behavior of propane as a major constituent of natural gases and water is crucial in the design and operation of natural gas and liquefied petroleum gas (LPG) facilities as well as reducing the risks associated with condensed water formation (e.g., ice, gas hydrates, and

corrosion) during transportation and processing of natural gases and LPG. Finally, hydrocarbon solubility in water is also an important issue from an environmental point of view due to new legislations and restrictions on the hydrocarbon content in water disposal.

Accurate solubility data, especially at low temperatures, are necessary to develop and verify thermodynamic models. However, preliminary studies show that most of the experimental data for hydrocarbons and for non-hydrocarbon gases (e.g., N₂, CO₂ and H₂S) have been reported at relatively high temperatures. The few data reported at low temperatures, below 300 K, and pressures above the atmospheric pressure are often scattered and certainly associated to large uncertainties that may lead consequently to inaccurate computing models [1].

In this work, new solubility measurements of propane in water in the liquid water–vapor (*L_w*–*V*) region have been gen-

Abbreviations: AAD, average absolute deviation $AAD = \frac{1}{N} \sum \left| \frac{x_{exp} - x_{cal}}{x_{exp}} \right|$; AD, absolute deviation $AD = \left| \frac{x_{exp} - x_{cal}}{x_{exp}} \right|$; BIP, binary interaction parameter; FID, flame ionization detector; FOB, objective function; ITS, international temperature scale; LPG, liquefied petroleum gas; NDD, non-density dependent mixing rules; PID, proportional integrator derivative controller; TCD, thermal conductivity detector; VPT-EoS, Valderrama modification of Patel–Teja equation of state; GC, gas chromatograph

* Corresponding author. Tel.: +33 1 64 69 49 65; fax: +33 1 64 69 49 68.
E-mail address: richon@paris.ensmp.fr (D. Richon).

Table 1

A list of experimental solubility data of propane in water at pressures over than atmospheric pressure

Reference	<i>T</i> (K)	<i>P</i> (MPa)
Wehe and McKetta [8]	344.26	0.514–1.247
Sanchez and Coll [9]	473.15–663.15	20–330
Kobayashi and Katz [10]	278.87–422.04	0.496–19.210
Azarnosh and McKetta [11]	288.71–410.93	0.101–3.528

erated over a wide temperature range, i.e., 277.62–368.16 K up to 3.915 MPa. The different isotherms presented herein were obtained using an apparatus based on a static-analytic method taking advantage of a ROLSITM capillary sampler [1–5].

The Valderrama modification of the Patel–Teja equation of state (VPT-EoS) [6] with non-density dependent mixing rules (NDD) [7] is used to model propane–water equilibrium. The binary interaction parameters (BIPs) between propane and water are tuned using the new experimental results on propane solubility data. Then, the new experimental data are compared with the results of the thermodynamic model and some literature data. In order to further evaluate the reliability of the model, some independent literature data on the water content of the vapor phase are compared with the predictions of the model. Predictions are found to be in good agreement with the experimental data demonstrating the reliability of the experimental techniques and the model used in this work.

2. Literature review

Only a few studies have been reported on the solubility of propane in water, perhaps due to the fact that the solubilities of hydrocarbons decreases with an increase in the number of carbon atoms; therefore, the amount of propane solubilized in water is indeed very small. Tables 1 and 2 give a list of some experimental data on propane solubility in water and water content of the vapor phase in the L_w – V region, respectively.

Table 2

Literature review of water content of the vapor phase in propane–water system [1]

Reference	<i>T</i> _{min} (K)	<i>T</i> _{max} (K)	<i>P</i> _{min} (MPa)	<i>P</i> _{max} (MPa)
Song and Kobayashi [12]	235.65	299.65	0.621	1.097
Song and Kobayashi [13]	235.65	299.65	0.621	1.097
Sloan et al. [14]	246.7	276.5	0.772	0.772
Bourrie and Sloan [15]	264.15	264.15	3.446	3.446
Sparks and Sloan [16]	246.66	276.43	0.772	0.772
Parrish et al. [17]	243.15	303.15	Slightly above <i>P</i> ^{sat}	Slightly above <i>P</i> ^{sat}
Pollin et al. [18] (internal report)	N/A	N/A	N/A	N/A
Kahre [19] (internal report)	N/A	N/A	N/A	N/A
Bukacek [20]	377.76	377.87	7.176	67.38
Kobayashi and Katz [10]	278.87	422.04	0.567	19.32
Kobayashi [21]	N/A	N/A	N/A	N/A
Poettmann and Dean [22]	288.71	359.26	N/A	N/A
Chaddock [23]	299.82	369.65	1.137	17.232
Perry [24]	310.93	310.93	1.351	1.351
Hachmuth [25]	310.93	310.93	1.351	1.351
McKetta and co-workers [26]	N/A	N/A	N/A	N/A

N/A: Not Available.

Although solubility measurement at atmospheric pressure is not the purpose of this paper, a suitable reference for such property is the data compilation of the IUPAC Solubilities Data Series (volume 24 for propane) [27] and the different correlations for Henry constant determination available in the literature [28–30].

3. Experimental

3.1. Materials

Propane was purchased from Messer Griesheim with a certified purity greater than 99.995 vol.%. Helium (carrier gas) from Air Liquide is pure grade with traces of water (3 ppm) and of hydrocarbons (0.5 ppm). Deionized water was used after degassing.

3.2. Apparatus and experimental procedures

The apparatus used in this work (Fig. 1) is based on a static-analytic method with fluid-phase sampling. This apparatus is similar to that described by Laugier and Richon [31].

The phase equilibrium is achieved in a cylindrical cell (equilibrium cell: EC) made of sapphire (sapphire tube: ST), the cell volume is about 28 cm³ (internal diameter = 25 mm) and it can be operated up to 8 MPa between 223.15 and 473.15 K. The cell is immersed in an ULTRA-KRYOMAT LAUDA constant-temperature liquid bath (LB) that controls and maintains the desired temperature within ±0.01 K. In order to perform accurate temperature measurements in the equilibrium cell and to check for thermal gradients, temperature is measured at two locations corresponding to the vapor and liquid phases through two 100 Ω platinum resistance thermometer devices (Pt 100) connected to a HP data acquisition unit (HP34970A). These two Pt 100 are carefully and periodically calibrated against a 25 Ω reference platinum resistance thermometer (TINSLEY precision instruments). The result-

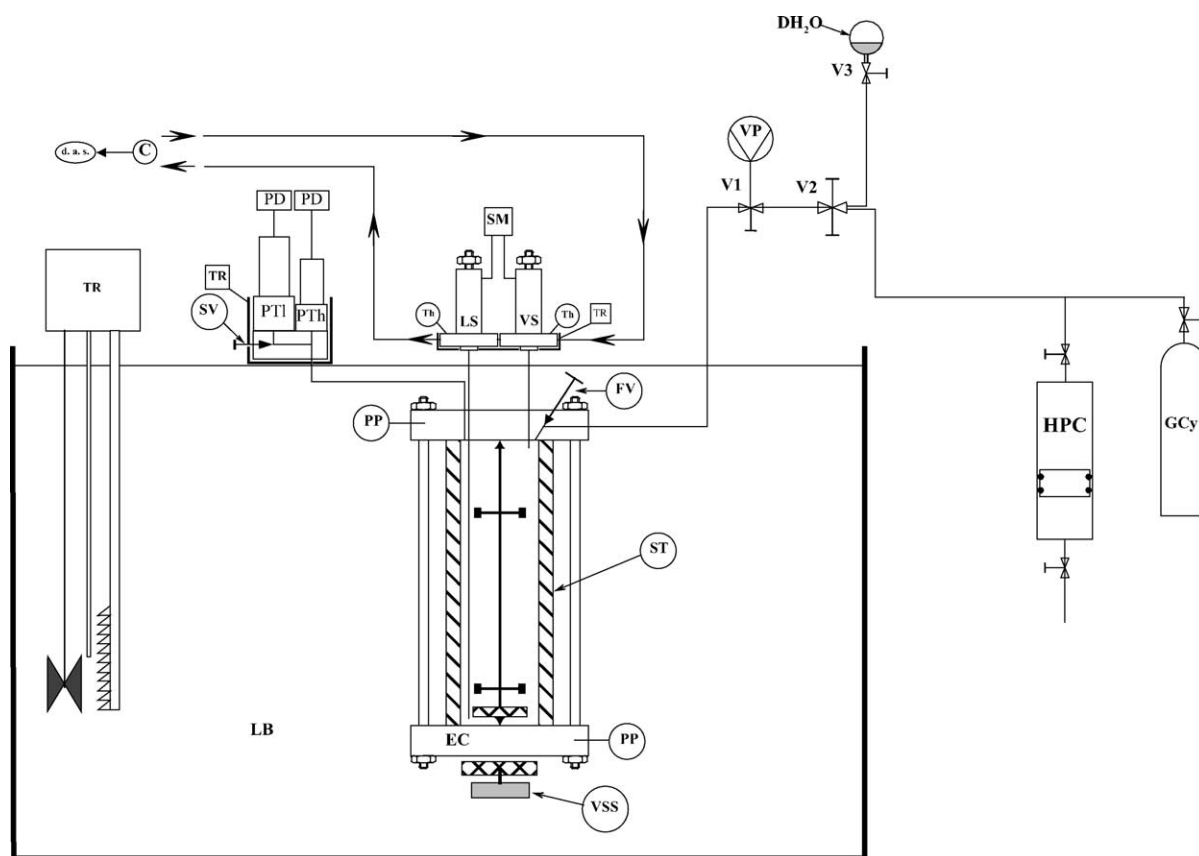


Fig. 1. Flow diagram of the equipment [1–5]: C: carrier gas; d.a.s.: data acquisition system; DH_2O : degassed water; EC: equilibrium cell; FV: feeding valve; GCy: gas cylinder; HPC: high-pressure compressor; LB: liquid bath; LS: liquid sampler; PD: pressure display; PP: platinum resistance thermometer probe; PTh and PTI: high- and low-pressure transducers; SM: sampler monitoring; ST: sapphire tube; SV: home-made shut off valve; Th: Thermocouple; TR: Temperature Regulator; Vi: valve i; VS: vapor sampler; VSS: Variable Speed Stirrer; VP: vacuum pump.

ing uncertainty is not higher than ± 0.04 K. The $25\ \Omega$ reference platinum resistance thermometer (PP) was calibrated by the Laboratoire National d'Essais (Paris) based on the 1990 International Temperature Scale (ITS 90). Pressures are measured by means of two Druck pressure transducers (one for 0–0.6 MPa, PTI and the other for pressures up to 6 MPa, PTh) connected to the HP data acquisition unit (HP34970A); the pressure transducers are maintained at constant temperature (temperature higher than the highest temperature of the study) by means of a specially made oven, which is controlled using a PID regulator (TR) (WEST, model 6100). Both pressure transducers are calibrated against a dead-weight pressure balance (Desgranges & Huot 5202S, CP 0.3–40 MPa, Aubervilliers, France). Pressure measurement uncertainties are estimated to be within ± 0.1 kPa in the 0–0.8 MPa range using the 0–0.6 MPa pressure transducer and ± 1 kPa in the 0.8–6 MPa range using the 0–6 MPa pressure transducer.

The HP on-line data acquisition unit is connected to a personal computer through an RS-232 interface. This system allows real time readings and storage of temperatures and pressures throughout the different isothermal runs.

The analytical work was carried out using a gas chromatograph (VARIAN model CP-3800) equipped with two detectors in series, a thermal conductivity detector (TCD)

and a flame ionization detector (FID), connected to a data acquisition system (BORWIN ver 1.5, from JMBS, Le Fontanil, France) (d.a.s.). The analytical column is a Haysep T 100/120 Mesh column (silcosteel tube, length: 1.5 m, diameter: 1/8 inch). The TCD was used to detect the water; it was repeatedly calibrated by injecting known amounts of water through liquid-tight syringes. The uncertainties on the calculated mole numbers of water are estimated to be within $\pm 1.5\%$ in the range of 2.5×10^{-5} to 3×10^{-4} water mole numbers. The FID was used to detect the hydrocarbon. It was first calibrated by introducing known amounts of nitrogen through a gas-tight syringe, the mole numbers of nitrogen are thus known within $\pm 1\%$. After this careful nitrogen calibration, different propane + nitrogen mixtures of known low-propane composition were prepared inside the equilibrium cell. Then, samples of different (a priori) unknown sizes are withdrawn directly from the cell through the ROLSITM sampler for GC analyses. Knowing the composition of the mixture and the number of sampled moles of nitrogen (through the response of the TCD), it is possible to estimate the amount of propane and hence calibrate the FID (Fig. 2) for this compound. A polynomial expression of the second order is adjusted to the propane calibration curve; this expression allows recalculating the injected moles of propane as a function of

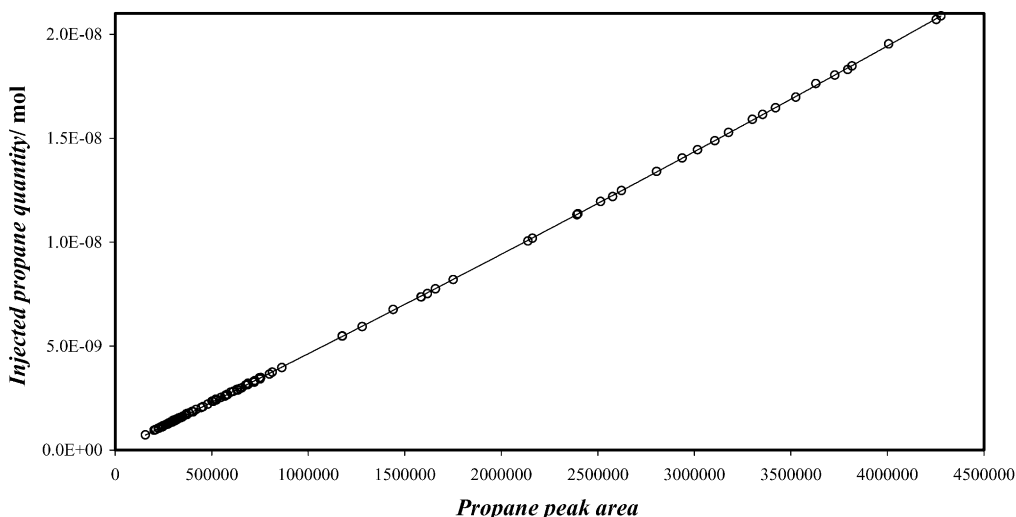


Fig. 2. FID calibration curve with Propane.

the FID response. The resulting relative uncertainty in the moles of propane is about $\pm 3\%$ (Fig. 3) below 5×10^{-9} and $\pm 1\%$ over 5×10^{-9} .

3.3. Experimental procedures

The equilibrium cell and its loading lines are evacuated down to 0.1 Pa before filling it partially (approximately 5 cm^3) with degassed water using the auxiliary cell. Then, successive amounts of propane are introduced into the cell directly from the commercial cylinder. After introduction of gas into the cell, efficient stirring is started and pressure is stabilized within a few minutes, measurements are performed only when pressure is constant within experimental uncertainty (furthermore pressure is ensured to remain constant in all sample analyses).

For each equilibrium condition after adding propane, at least 10 samples of the liquid phase are withdrawn using the pneumatic ROLSITM samplers [32] and analyzed in order to check for measurement repeatability. As the volume of the withdrawn samples (typically less than $1 \mu\text{l}$) is very small compared to the volume of the liquid phase present in the equilibrium cell (more than $5000 \mu\text{l}$), it is possible to withdraw many samples without disturbing the phase equilibrium.

4. Thermodynamic model

4.1. Pure compound properties

The critical temperature (T_c), critical pressure (P_c), critical volume (v_c), and acentric factor (ω), for each of the two pure compounds are provided in Table 3 [33].

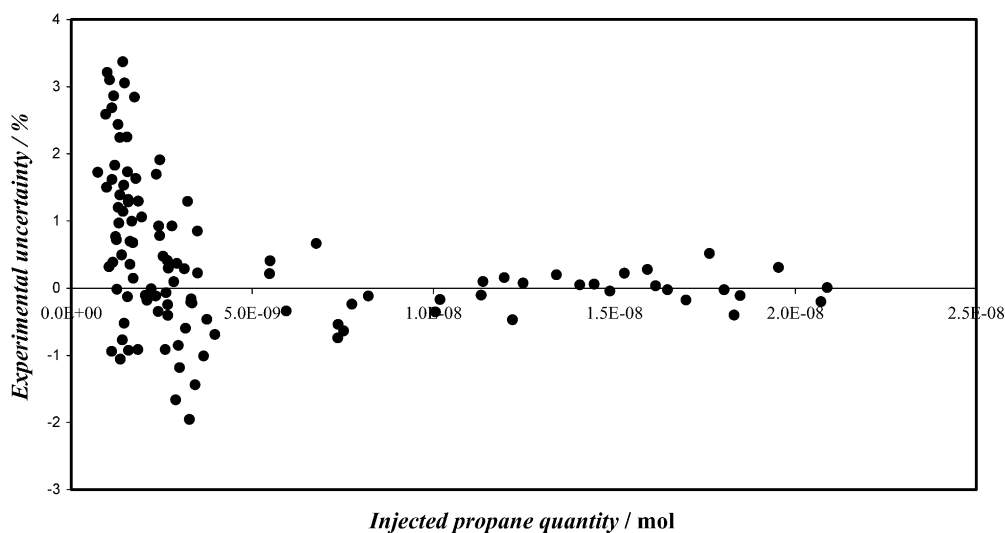


Fig. 3. Propane calibration; deviation plot.

Table 3
Critical properties and acentric factors (from Avlonitis [33])

Compound	P_c (MPa)	T_c (K)	v_c (m ³ kmol ⁻¹)	ω
Water	22.048	647.30	0.0560	0.3442
Propane	4.2496	369.82	0.2029	0.15176

4.2. Model

A general phase equilibrium model based on uniformity of the fugacity of each component throughout all the phases [34,35] was used to model the gas solubility (i.e., L_w –V equilibrium). The VPT-EoS [6] with the NDD mixing rules [7] was employed in calculating fugacities in the fluid phases. This combination has proved to be a strong tool in modeling systems with polar as well as non-polar components [7].

The VPT-EoS [6] is given by:

$$P = \frac{RT}{v-b} - \frac{\bar{a}\alpha(T_r)}{v(v+b) + c(v-b)} \quad (1)$$

with

$$\bar{a} = \frac{\Omega_a R^2 T_c^2}{P_c} \quad (2)$$

$$b = \frac{\Omega_b R T_c}{P_c} \quad (3)$$

$$c = \frac{\Omega_{c^*} R T_c}{P_c} \quad (4)$$

$$\alpha(T_r) = [1 + F(1 - T_r^\Psi)]^2 \quad (5)$$

where P is the pressure, T the temperature, v the molar volume, R the universal gas constant, and $\Psi = 0.5$. The subscripts c and r denote critical and reduced properties, respectively.

The coefficients Ω_a , Ω_b , Ω_{c^*} , and F are given by:

$$\Omega_a = 0.66121 - 0.76105 Z_c \quad (6)$$

$$\Omega_b = 0.02207 + 0.20868 Z_c \quad (7)$$

$$\Omega_{c^*} = 0.57765 - 1.87080 Z_c \quad (8)$$

$$F = 0.46283 + 3.58230(\omega Z_c) + 8.19417(\omega Z_c)^2 \quad (9)$$

where Z_c is the critical compressibility factor, and ω the acentric factor. Tohidi-Kalorazi [36] relaxed the α function for water, α_w , using experimental water vapor-pressure data in the range of 258.15–374.15 K, in order to improve the predicted water fugacity:

$$\alpha_w(T_r) = 2.4968 - 3.0661 T_r + 2.7048 T_r^2 - 1.2219 T_r^3 \quad (10)$$

Eq. (10) is used in the present work.

In this work, the NDD mixing rules developed by Avlonitis et al. [7], are used to calculate the a -parameter in mixtures

Table 4

Experimental and calculated propane mole fractions in the aqueous phase of the propane–water system

T (K)	P_{exp} (MPa)	$x_{\text{exp}} \times 10^4$	AAD (exp) (%)	$x_{\text{cal}} \times 10^4$	AD (%)
277.62	0.378	2.235	2.6	1.946	12.93
278.09	0.357	2.061	0.9	1.816	11.89
278.09	0.357	2.107	2.1	1.816	13.81
278.09	0.398	2.208	2.9	2.005	9.19
278.09	0.445	2.439	1.0	2.218	9.06
280.14	0.395	2.027	1.8	1.851	8.68
280.14	0.447	2.245	2.9	2.070	7.80
280.14	0.504	2.461	1.3	2.304	6.38
280.14	0.557	2.694	2.7	2.515	6.64
283.06	0.401	1.796	2.7	1.696	5.57
283.06	0.46	2.028	2.0	1.921	5.28
283.06	0.522	2.266	0.5	2.150	5.12
283.06	0.612	2.555	0.5	2.470	3.33
288.13	0.488	1.733	3.7	1.718	0.87
288.13	0.697	2.347	2.4	2.346	0.04
288.13	0.596	2.080	1.3	2.051	1.39
288.13	0.39	1.423	1.1	1.401	1.55
293.13	0.399	1.229	0.7	1.229	0.00
293.13	0.8	2.249	3.8	2.272	1.02
293.13	0.699	2.023	1.0	2.028	0.25
293.13	0.599	1.756	2.0	1.774	1.03
293.13	0.5	1.510	2.8	1.510	0.00
298.12	0.401	1.037	1.0	1.071	3.28
298.12	0.493	1.244	1.3	1.296	4.18
298.12	0.58	1.438	1.3	1.500	4.31
298.12	0.675	1.650	0.5	1.714	3.88
298.12	0.812	1.938	0.2	2.007	3.56
298.12	0.92	2.144	1.6	2.224	3.73
308.13	0.399	0.770	1.3	0.825	7.14
308.13	0.61	1.158	2.7	1.221	5.44
308.13	0.848	1.525	2.1	1.629	6.82
308.13	1.003	1.755	0.9	1.873	6.72
308.13	1.191	2.007	0.4	2.146	6.93
323.13	0.425	0.598	1.6	0.635	6.19
323.13	0.65	0.897	3.2	0.947	5.57
323.13	0.898	1.199	2.7	1.263	5.34
323.13	1.156	1.485	1.4	1.563	5.25
323.13	1.396	1.728	0.7	1.815	5.03
323.13	1.665	1.957	0.8	2.066	5.57
338.15	0.403	0.449	1.0	0.461	2.67
338.15	0.697	0.766	1.3	0.787	2.74
338.15	0.997	1.094	2.3	1.091	0.27
338.15	1.302	1.370	1.1	1.370	0.00
338.15	1.632	1.621	0.7	1.640	1.17
338.15	2.008	1.899	1.3	1.907	0.42
338.15	2.292	2.082	0.3	2.080	0.10
353.18	0.404	0.368	0.2	0.366	0.54
353.18	0.696	0.646	0.3	0.642	0.62
353.18	0.972	0.917	0.0	0.884	3.60
353.18	1.431	1.317	1.5	1.246	5.39
353.18	2.061	1.745	0.6	1.665	4.58
353.18	2.483	2.006	1.0	1.897	5.43
353.18	3.082	2.270	1.4	2.158	4.93
368.16	0.41	0.321	0.3	0.296	7.79
368.16	1.028	0.891	0.4	0.803	9.88
368.16	1.433	1.203	0.2	1.095	8.98
368.16	1.94	1.586	0.6	1.419	10.53
368.16	2.495	1.989	0.7	1.723	13.37
368.16	2.997	2.236	2.5	1.952	12.70
368.16	3.503	2.482	0.7	2.141	13.74
368.16	3.915	2.601	0.7	2.262	13.03

AAD (exp): Average absolute deviation between samples AAD(exp) = $\frac{1}{N} \sum \left| \frac{x_{\text{exp}} - x_{\text{cal}}}{x_{\text{exp}}} \right|$ and AD: Absolute deviation AD = $\left| \frac{x_{\text{exp}} - x_{\text{cal}}}{x_{\text{exp}}} \right|$.

consisting of polar and non-polar compounds:

$$a = \bar{a}\alpha(T_r) = a^C + a^A \quad (11)$$

where a^C is given by the classical quadratic mixing rules as follows:

$$a^C = \sum_i \sum_j x_i x_j a_{ij} \quad (12)$$

and b , c and a_{ij} parameters are expressed by:

$$b = \sum_i x_i b_i \quad (13)$$

$$c = \sum_i x_i c_i \quad (14)$$

$$a_{ij} = (1 - k_{ij})\sqrt{a_i a_j} \quad (15)$$

where k_{ij} is the standard BIP.

The term a^A corrects for asymmetric interaction, which cannot be efficiently accounted for by classical mixing rules:

$$a^A = \sum_p x_p^2 \sum_{i \neq p} x_i a_{pi} l_{pi} \quad (16)$$

$$a_{pi} = \sqrt{a_p a_i} \quad (17)$$

$$l_{pi} = l_{pi}^0 - l_{pi}^1(T - T_0) \quad (18)$$

where p is the index of polar components and T_0 is equal to 273.15 K.

5. Results and discussions

The experimental and calculated gas solubility data are reported in Table 4 and Fig. 4.

Table 5

BIPs between hydrocarbon (propane), hc, and water, w, for the VPT-EoS [6] and NDD mixing rules [7]

BIP	Value
k_{w-hc}	0.65117
l_{w-hc}^0	1.81373
$l_{w-hc}^1 \times 10^4$	37.70874

As mentioned before, the BIPs between propane–water are adjusted directly to the measured propane solubility data through a Simplex algorithm using the following objective function, FOB:

$$FOB = \frac{1}{N} \sum_1^N \left| \frac{x_{i,exp} - x_{i,cal}}{x_{i,exp}} \right| \quad (19)$$

where N is the number of data points, $x_{i,exp}$ the measured property, and $x_{i,cal}$ the calculated value. Table 5 reports the optimized BIPs values between propane and water used in this work.

Our isothermal P , x data sets for the propane–water are well represented with the VPT-EoS [6] and NDD mixing rules [7] (AAD = 5.3%). Propane solubility data reported in the literature are plotted in Fig. 5. As it can be seen, the data reported in the literature are fairly dispersed and to some extent scattered, particularly the data of Azarnooosh and McKetta [11] at 288.71 K. Henry's constants for propane in water (H_{iw}) have been derived from our data adjusting parameters of Eq. (20):

$$\ln(H_{iw}) \text{ (kPa)} = 552.64799 + 0.078453T - \frac{21334.4}{T} - 85.89736 \ln T \quad (20)$$

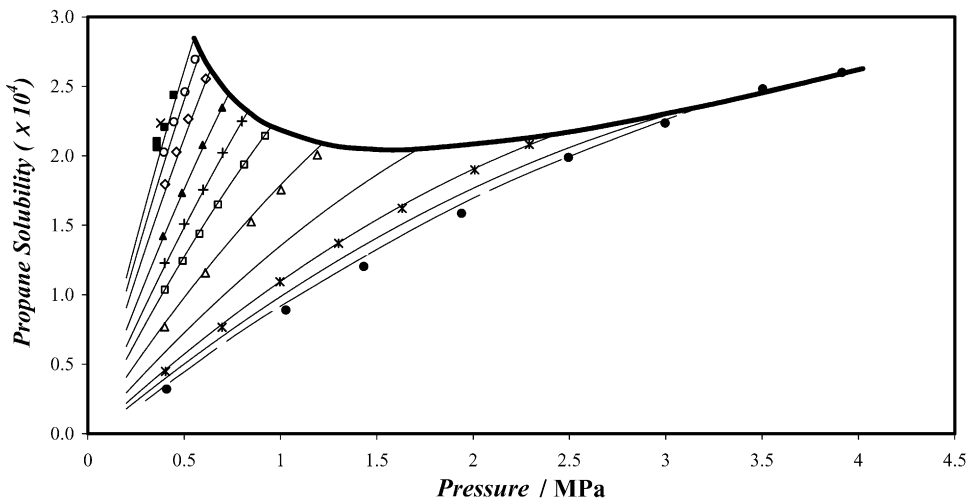


Fig. 4. Propane mole fraction in the aqueous phase as a function of pressure at various temperatures. (x) 277.62 K; (■) 278.09 K; (○) 280.14 K; (◇) 283.06 K; (▲) 288.13 K; (+) 293.13 K; (□) 298.12 K; (△) 308.13 K; (●) 323.13 K; (*) 338.15 K; (■) 353.18 K; (●) 368.16 K. Thin solid lines, calculated with the VPT-EoS [6] and NDD mixing rules [7] with parameters from Table 5, at the above experimental temperatures. Bold solid line: VLL three-phase line, delimiting the vapor–liquid and liquid–liquid co-existence regions.

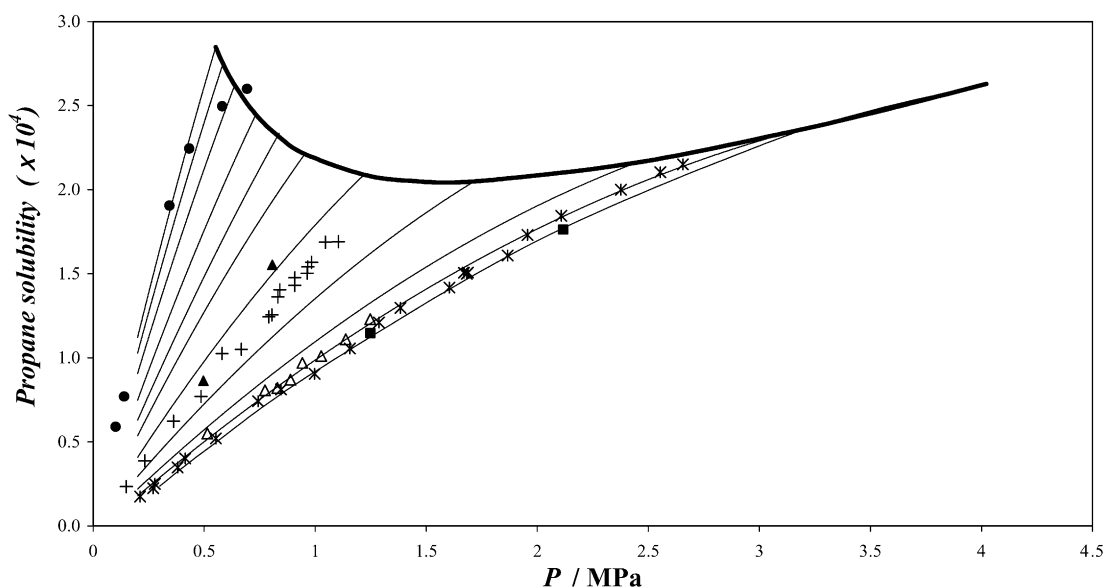


Fig. 5. Comparison of experimental propane solubilities in water. (●) 288.71 K from Azarnoosh and McKetta [11]; (▲) 310.93 K from Kobayashi and Katz [10]; + 310.93 from Azarnoosh and McKetta [11]; (Δ) 344.26 K from Wehe and McKetta [8]; (×) 344.26 K from Azarnoosh and McKetta [11]; (■) 349.82 K from Kobayashi and Katz [10]. Solid lines, calculated with the VPT-EoS [6] and NDD mixing rules [7] with parameters from Table 5 at 278.09, 280.14; 283.06, 288.13, 293.13, 298.12, 308.13, 323.13, 338.15, 353.18, and 368.16 K (from left to right). Bold solid line: VLL three-phase line, delimiting the vapor–liquid and liquid–liquid co-existence regions.

Carroll and Mather [29] reported the following Henry's constant correlation: Eq. (21) that has parameters very similar to our's (Eq. (20)).

$$\ln(H_{iw}) \text{ (kPa)} = 552.65815 + 0.077514T - \frac{21334.4}{T} - 85.84949 \ln T \quad (21)$$

Our isothermal P, x data sets for the propane–water are well represented with Eq. (20) (AAD = 1.5%) while AAD = 2.7% with Eq. (21). In order to further evaluate the performance

of the model, some experimental water content data in the literature are compared with the predictions of the model using the BIPs adjusted on the solubility data generated in this work. Table 2 shows a list of experimental water content data for water–propane system [1]. As it can be seen, most of these data have been reported at relatively high temperatures. Table 6 shows a comparison between experimental water content data and the predictions through the model. Predictions show satisfactory agreement with these experimental data demonstrating the reliability of model used in this work.

Table 6
Experimental and calculated water mole fractions in the vapor phase of the propane–water system

T (K)	P_{exp} (MPa)	Experimental water content	Calculated water content	AD (%)
Kobayashi and Katz [10]				
310.93	0.703	0.009540	0.008905	6.66
310.93	0.972	0.006960	0.006501	6.59
338.71	1.006	0.026490	0.024450	7.70
338.71	1.496	0.017020	0.015790	7.23
338.71	1.992	0.012000	0.011190	6.75
360.93	1.310	0.049670	0.047690	3.99
360.93	2.130	0.029100	0.027660	4.95
360.93	3.205	0.017220	0.015900	7.67
Song and Kobayashi [13]				
281.95	0.621	0.002000	0.001718	14.10
285.25	0.679	0.002028	0.001950	3.85
292.65	0.819	0.002668	0.002568	3.75
299.65	0.897	0.003494	0.003576	2.35

6. Conclusions

New experimental data on the solubility of propane in water (in the L_w –V region) were generated at low temperatures (i.e., 277.62–368.16 K) up to 3.915 MPa, using a static-analytic apparatus, taking advantage of a ROLSI™ high-pressure capillary sampler.

The Valderrama modification of the Patel–Teja equation of state with non-density dependent mixing rules is employed to model the phase behavior in this system. The new experimental data have been used in order to tune the BIPs between propane and water. Using these tuned binary interaction parameters, the predictions of the model have been compared to independent gas solubilities and water content data from the literature. The predictions are in good agreement with the experimental data demonstrating the reliability of the model used in this work.

List of symbols

a	attractive parameter of the equation of state
$\bar{a}$	parameter of the equation of state
b	parameter of the equation of state
c	parameter of the equation of state
F	parameter of the equation of state
H	Henry's constant
k	binary interaction parameter for the classical mixing rules
l	binary interaction parameter for the asymmetric term
L	liquid
N	number of experimental points
P	pressure (Pa)
R	universal gas constant (J/(mol K))
T	temperature (K)
v	molar volume
V	vapor
x	mole fraction
$\bar{x}$	average mole fraction
Z	compressibility factor

Greek letters

Ψ	power parameter in the VPT-EoS
Ω	parameter in the VPT-EoS
α	temperature dependent function
ω	acentric factor

Superscript

A	asymmetric properties
C	classical Properties
sat	saturation Condition
0	non-temperature dependent term in NDD mixing rules
1	temperature dependent term in NDD mixing rules

Subscripts

cal	calculated property
exp	experimental property
hc	hydrocarbon
max	maximum
min	minimum
a	related to the a parameter of the VPT-EoS
b	related to the b parameter of the VPT-EoS
c	critical property
c^*	related to the c parameter of the VPT-EoS
i, j	molecular species
p	polar compound
r	reduced properties
w	water
0	reference property

Acknowledgement

The financial support by the European Infrastructure for Energy Reserve Optimization (EIERO) that has provided

the opportunity for this joint work is gratefully acknowledged.

References

- [1] A.H. Mohammadi, A. Chapoy, D. Richon, B. Tohidi, Ind. Eng. Chem. Res. (2004), in press.
- [2] A. Chapoy, A.H. Mohammadi, D. Richon, B. Tohidi, Fluid Phase Equilib. 220 (2004) 113–121.
- [3] A. Chapoy, A.H. Mohammadi, B. Tohidi, D. Richon, J. Chem. Eng. Data 49 (4) (2004) 1110–1115.
- [4] A. Chapoy, C. Coquelet, D. Richon, Fluid Phase Equilib. 214 (2003) 101–117.
- [5] A. Chapoy, C. Coquelet, D. Richon, J. Chem. Eng. Data 48 (2003) 957–966.
- [6] J.O. Valderrama, J. Chem. Eng. Jpn. 23/1 (1990) 87–91.
- [7] D. Avlonitis, A. Danesh, A.C. Todd, Fluid Phase Equilib. 94 (1994) 181–216.
- [8] A.H. Wehe, J.J. McKetta Jr., Anal. Chem. 33/2 (1961) 291–293.
- [9] M. Sanchez, R. Coll, An. Quim. 74 (1978) 132.
- [10] R. Kobayashi, D.L. Katz, Ind. Eng. Chem. Res. 45/2 (February 1953) 440–451.
- [11] A. Azarnoosh, J.J. McKetta Jr., Pet. Refiner. 37 (1958) 275–278.
- [12] K. Y. Song, R. Kobayashi, Research Report (RR-132), GPA, Tulsa, OK, 1991.
- [13] K.Y. Song, R. Kobayashi, Fluid Phase Equilib. 95 (1994) 281–298.
- [14] E.D. Sloan, K.A. Sparks, J.J. Johnson, Ind. Eng. Chem. Res. 26 (1987) 1173–1179.
- [15] M.S. Bourrie, E.D. Sloan, Research Report (RR-100), GPA, Tulsa, OK, June 1986.
- [16] K.A. Sparks, E.D. Sloan, Research Report (RR-71), GPA, Tulsa, OK, 1983.
- [17] W.R. Parrish, A.G. Pollin, T.W. Schmidt, Proceedings of the 61st Ann. Conv. Gas Process Assoc., Dallas, TX, 1982.
- [18] Pollin et al., Internal report, Phillips Petroleum Co., 1982 (cited in ref. 11).
- [19] L.C. Kahre, Internal report, Phillips Petroleum Co., 1964 (and also, L.C. Kahre, Internal report, Phillips Petroleum Co., Bartlesville, OK, 1968).
- [20] R. F. Bukacek, Institute of Gas Technology, Research Bulletin, vol. 8, 1955.
- [21] R. Kobayashi, Ph.D. Thesis, University of Michigan, 1951.
- [22] F.H. Poettmann, M.R. Dean, Pet. Refiner. 25/12 (December 1946) 125–128.
- [23] Chaddock, Ph.D. Dissertation, University of Michigan, 1940.
- [24] C.W. Perry, Ind. Eng. Chem. Anal. Ed. 10/9 (1938) 513–514.
- [25] K.H. Hachmuth, Western Gas 8/1 (1931) 55.
- [26] J.J. McKetta Jr., co-workers, unpublished data (cited in: W.F. Hoot, A. Azarnoosh, J.J. McKetta, Pet. Refiner. 36/5 (May 1957) 255–256).
- [27] IUPAC Solubility Data Series, vol. 24, Pergamon Press, 1986.
- [28] Y.K. Li, L.X. Nghiem, Can. J. Chem. Eng. 64 (1986) 486–496.
- [29] J.J. Carroll, A.E. Mather, Chem. Eng. Sci. 52 (1997) 545–552.
- [30] C.L. Yaws, J.R. Hopper, X. Wang, A.K. Rathinsamy, R.W. Pike, Chem. Eng. J. (1999) 102–105.
- [31] S. Laugier, D. Richon, Rev. Sci. Instrum. 57 (1986) 469–472.
- [32] P. Guilbot, A. Valtz, H. Legendre, D. Richon, Analisis 28 (2000) 426–431.
- [33] D.A. Avlonitis, M.Sc. Thesis, Heriot-Watt University, 1988.
- [34] D.A. Avlonitis, Ph.D. Thesis, Heriot-Watt University, 1992.
- [35] B. Tohidi, R.W. Burgass, A. Danesh, A.C. Todd, Proceedings of the SPE Offshore Europe 93 Conference, SPE 26701, 1993, pp. 255–264.
- [36] B. Tohidi-Kalorazi, Ph.D. Thesis, Heriot-Watt University, 1995.